Band-1 Computational Experiment: Chromatic Number of the Plane — Moser Spindle Verification and Bound Ledger

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2026-09-11

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Honesty stamp. This note does **not** claim that $\chi(\mathbb{R}^2)$ is solved, does **not** claim a new numeric bound, and does **not** stamp Clay-closed or a finished Theory of Everything. The standing published interval remains

$$5 \leq \chi(\mathbb{R}^2) \leq 7.$$

Tonight we independently re-verify one classical finite unit-distance obstruction (the Moser spindle: $\chi = 4$) and record the literature status of de Grey's 2018 five-chromatic unit-distance graphs that raised the lower bound from 4 to 5.

Abstract

We execute tonight's Band-1 collapse item 5 (chromatic-number-of-the-plane graph hunt). On this Linux box we:

1. Embed the **Moser spindle** (7 vertices, 11 edges) in $\mathbb{R}^2$ via the Moser-lattice coordinates $\omega_1 = \frac{1}{2} + i\frac{\sqrt{3}}{2}$, $\omega_3 = \frac{5}{6} + i\frac{\sqrt{11}}{6}$, and the vertex set
$$M_7 = \{0,\ 1,\ \omega_1,\ \omega_3,\ \omega_1\omega_3,\ 1+\omega_1,\ \omega_3+\omega_1\omega_3\}.$$
2. Certify that exactly **11** pairwise distances equal 1 (tolerance 10^{-9}; algebraic identities confirm key edges exactly).
3. Exhaustively certify **not 3-colorable** and **4-colorable** (backtrack + 3^6 enumeration with vertex-0 color fixed): $\chi(M_7) = 4$.
4. Document de Grey (2018) and subsequent reductions as the literature source of $\chi(\mathbb{R}^2) \geq 5$, without claiming a fresh SAT certificate of a 5-chromatic unit-distance graph in this run.

Bound move this run: none. Hygiene / methods-validation only. Interval unchanged: $5 \leq \chi \leq 7$.

Problem statement

The **Hadwiger–Nelson problem** asks for the chromatic number of the plane $\chi(\mathbb{R}^2)$: the least number of colors needed to color the Euclidean plane so that no two points at distance 1 share a

color. Equivalently, it is the chromatic number of the infinite unit-distance graph G_∞ with $V = \mathbb{R}^2$ and edges all pairs at Euclidean distance 1.

By the de Bruijn–Erdős theorem (assuming the axiom of choice), $\chi(\mathbb{R}^2)$ equals the supremum of $\chi(H)$ over finite unit-distance subgraphs $H \subseteq G_\infty$. Finite graph hunts therefore move the lower bound; tiling / hexagonal colorings move the upper bound.

Bound	Value	Mechanism
Lower (classical)	≥ 4	Moser spindle (1961); also Golomb graph
Lower (current)	≥ 5	de Grey (2018) and shrinks (Heule, etc.)
Upper (current)	≤ 7	hexagonal tiling / 7-coloring of the plane

Tonight's item asks for one concrete unit-distance graph attempt that would raise 5 or lower 7. Raising above 5 or lowering below 7 is a research-scale attack. We therefore **verify the classical $\chi = 4$ spindle computationally** and **ledger the $\chi \geq 5$ literature**, documenting limits.

Method

Moser spindle (primary computational certificate)

Construction (Moser–Moser 1961; lattice form as in arXiv:2311.10069, Example 1):

- Two unit rhombi with angles 60°/120° (each containing two equilateral triangles).
- Share one acute vertex at the origin of the spindle assembly.
- Relative rotation chosen so the remaining two acute vertices are themselves at unit distance (the "spindle tip" edge that forces the fourth color).

Algebraic embedding used here:

$$\omega_1 = e^{i\pi/3} = \tfrac{1}{2} + i\tfrac{\sqrt{3}}{2}, \qquad \omega_3 = \tfrac{5}{6} + i\tfrac{\sqrt{11}}{6}.$$

Identity check: $|\omega_1| = 1$, $|\omega_3|^2 = (25 + 11)/36 = 1$. Tip–tip distance:

$$\big|(1 + \omega_1) - (\omega_3 + \omega_1\omega_3)\big| = |1 + \omega_1|\,|1 - \omega_3| = \sqrt{3} \cdot \frac{\sqrt{3}}{3} = 1.$$

Implementation: pure Python 3 (`verify_moser_spindle.py`), `math` only. Unit edges collected at absolute tolerance 10^{-9}. Chromatic number by sequential-coloring backtrack and by exhaustive product enumeration with v_0 color-fixed.

de Grey 2018 (literature ledger, not re-SAT-certified tonight)

Aubrey de Grey (2018) exhibited a finite unit-distance graph with chromatic number 5 (original order on the order of 1581 vertices), raising $\chi(\mathbb{R}^2)$ from 4 to 5. Subsequent computer-assisted shrinks (notably Marijn Heule and collaborators) produced much smaller 5-chromatic unit-distance graphs (orders in the low hundreds / below). A full independent SAT / DRAT replay of a modern shrink is a valuable next experiment but exceeds tonight's single-unit hygiene budget; we do **not** claim to have re-certified $\chi \geq 5$ from raw clauses in this run.

Upper bound (ledger only)

The classical hexagonal mosaic yields a 7-coloring of the plane in which each color class avoids distance 1, hence $\chi(\mathbb{R}^2) \leq 7$. No attempt was made tonight to construct a 6-coloring or to refute one.

Experiment design

Parameter	Choice
Probe	Moser spindle unit-distance + $\chi = 4$ certificate; de Grey literature ledger
Slug	`chromatic_plane_graph_hunt`
Host	Linux box, Python 3.13
Success criterion	Assert M_7 has 11 unit edges and $\chi = 4$; record $5 \leq \chi(\mathbb{R}^2) \leq 7$; no overclaim
Wall budget	$\ll 1$ second for the spindle

Code path: `/workspace/experiments/band1/chromatic_plane_graph_hunt/`.

Results (numbers!)

Moser spindle certificate

Check	Result
Vertices $\|V\|$	**7**
Unit edges $\|E\|$ (tol 10^{-9})	**11**
Degree sequence	$[4, 3, 3, 3, 3, 3, 3]$
3-colorable?	**No** ($3^6 = 729$ colorings with v_0 fixed; 0 proper)
4-colorable?	**Yes** (witness coloring $(0, 1, 2, 1, 2, 0, 3)$)
$\chi(M_7)$	**4**
Wall time	≈ 0.001 **s**

Unit edge list (vertex indices 0..6 in the M_7 order above):

$$\{0{-}1,\ 0{-}2,\ 0{-}3,\ 0{-}4,\ 1{-}2,\ 1{-}5,\ 2{-}5,\ 3{-}4,\ 3{-}6,\ 4{-}6,\ 5{-}6\}.$$

Key floating-point distance checks (all equal to 1 within 10^{-12} except the non-edge $|w_1 w_3 - w_1| \approx 1/\sqrt{3}$):

Quantity	Value	unit?
$\|\omega_1\|$	1	yes
$\|\omega_3\|$	1	yes
$\|\omega_1 - 1\|$	1	yes
$\|(1 + \omega_1) - \omega_1\|$	1	yes
$\|\omega_1 \omega_3 - \omega_3\|$	1	yes

Quantity	Value	unit?
$\|(\omega_3 + \omega_1\omega_3) - \omega_3\|$	1	yes
tip–tip $\|(1 + \omega_1) - (\omega_3 + \omega_1\omega_3)\|$	1	yes

Standing bound ledger for $\chi(\mathbb{R}^2)$

Bound	Value	Source	This run
Lower	≥ 4	Moser spindle (1961)	**Independently re-verified**
Lower	≥ 5	de Grey (2018); Heule et al. shrinks	Literature only (not re-SAT)
Upper	≤ 7	Hexagonal 7-coloring	Literature only
Exact	unknown $\in \{5, 6, 7\}$	—	**No claim**

This experiment adds no new numeric bound.

Limitations

- The Moser spindle certificate is classical; its value tonight is methods validation and an executable reproducibility artifact, not a new theorem.
- Floating-point distances use IEEE-754 `float` with tolerance 10^{-9}; the critical identities ($|\omega_3| = 1$, tip–tip $= 1$) also hold exactly in $\mathbb{Q}(\sqrt{3}, \sqrt{11})$.
- We did **not** independently SAT-certify a de Grey / Heule 5-chromatic unit-distance graph.
- We did **not** attempt a 6-coloring of the plane or a new finite obstruction for $\chi \geq 6$.
- No claim that $\chi(\mathbb{R}^2)$ is determined; no Clay / ToE stamp.

Next experiment

1. **Raise lower bound hygiene:** import a published small 5-chromatic unit-distance graph (Heule shrink), verify unit distances algebraically or with interval arithmetic, and replay a SAT/DRAT non-4-colorability certificate.
2. **Upper-bound attack:** study obstacles to 6-colorings (tile defects, computer search for critical unit-distance subgraphs that block known 6-color candidates).
3. **Golomb graph:** independent 10-vertex $\chi = 4$ unit-distance embedding as a second classical certificate.
4. **Moser lattice search:** bounded enumeration of small subsets of the Moser lattice for candidate 5-chromatic seeds (research-scale; not claimed here).

Conclusion

Band-1 item 5 complete as a **hygiene / verification** unit: Moser spindle independently confirmed as a 7-vertex, 11-edge unit-distance graph with $\chi = 4$; de Grey line recorded as the literature source of $\chi \geq 5$; upper bound ≤ 7 unchanged. Standing interval:

$$5 \leq \chi(\mathbb{R}^2) \leq 7.$$

No new bound. No Clay-closed claim. No finished-ToE stamp.

References (selected)

- L. Moser, W. Moser (1961): spindle establishing $\chi(\mathbb{R}^2) \geq 4$.
- A. de Grey (2018): first published 5-chromatic unit-distance graph.
- M. J. H. Heule et al.: subsequent shrinks / SAT certificates for 5-chromatic unit-distance graphs.
- Hadwiger–Nelson problem surveys; hexagonal 7-coloring upper bound (classical).
- arXiv:2311.10069 (fractional chromatic number of the plane ≥ 4; Moser-lattice M_7 presentation used above).

ZoraASI / C.M. Baird — Band-1 tonight #5 — 2026-09-11